

FluxTangible: Simple and Dynamic Haptic Tangible with Bumps and Vibrations

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ABSTRACT

To add haptic feedback and tactile cues of a physical interface to interaction with mobile touch displays, several passive systems have been proposed as Tangible User Interfaces (TUIs). The haptic feedback from such passive objects can be customized by changing their shape and structure but cannot be changed dynamically. Therefore, we propose a simple TUI that can provide dynamic haptic feedback on a touch device. We use an electromagnet embedded in the tangible object and a magnetic sheet fixed on the back of a touch device. An electric current flowing through the electromagnet and the customizable magnetic pattern printed on the magnetic sheet together produce magnetic interference. The magnetic interference caused by these magnetic objects is actively generated during interaction with the touch device to provide dynamic haptic feedback of both bumps and vibrations.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Haptic devices*; *Touch screens*.

KEYWORDS

Tangible interactions, tactile, haptic, magnet, touchscreen

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1 INTRODUCTION

Mobile devices such as smartphones and tablets are highly portable, allowing people to access information anywhere and at any time. Although their touchscreens are often criticized for their lack of input/output expressiveness [11], commercially available smartphones are increasingly using built-in transducers to provide tactile feedback¹. However, the transducers' vibration sensation cannot replicate all of the haptic cues typically provided by physical interfaces. Therefore, in the field of Human-Computer Interaction (HCI),

¹Examples include Apple Inc.'s Taptic engine and Haptic touch.

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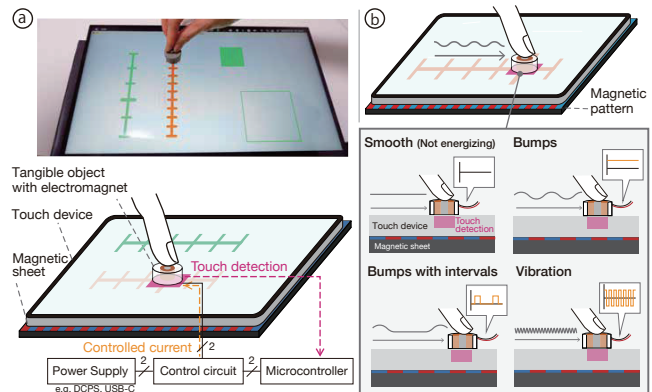


Figure 1: (a) System configuration comprising a magnetic sheet, a touch device, and a single tangible module equipped with an electromagnet. (b) Mechanism of our system. The haptic feedback to be presented is a range of bumps and vibrations only when the electromagnet is energized.

a variety of methods using passive objects have been proposed as Tangible User Interfaces (TUIs) [4] for touch displays [1, 5]. In these studies, some TUIs can present haptic feedback such as clicky and bumpy feelings using the shape and physical properties of the object [2, 7, 15] while others use magnetic interference [9, 14]. These methods enable users to design TUIs with haptic feedback in a simple and low-cost manner. The haptic feedback from these passive objects can be customized by changing their shape, structure, and magnetic field arrangement, but it cannot be changed dynamically. There are also TUIs that enable versatile haptic interactions with real-time control of magnetic fields [6, 10, 12]. Dynamic haptic feedback such as attraction, repulsion, and vibration can be provided by using the magnetic force between an object held by the user and the information presentation medium. However, such systems are not suitable for use with mobile touchscreens due to their bulkiness and weight – they are desktop-sized or large-scale devices containing 2D arrays of electromagnets. Thus there is a trade-off to be made between a large and complex system for highly dynamic haptic interaction and a small and simple system that can only provide a passive haptic presentation. Our resolution of the trade-off consists in a combination of the simple method using magnetic material with the more complex dynamic magnetic control technique to build a simple TUI that can dynamically vary its haptic feedback (see figure 1).

2 FLUXTANGIBLE

We utilize magnetic interference between a magnetic sheet attached to the back of a touch device and a small electromagnet used on the touchscreen of the device to present haptic feedback. Since recent commercially available touch devices are usually quite thin, magnetic forces from below the device pass right through it, enabling the haptic feedback on the screen. The magnetic field pattern on the magnetic sheet can be easily changed with a handheld magnetizing module, allowing us to design haptic presentations both simply and quickly. In addition, the use of an electromagnet allows multiple haptic feedback to be presented at arbitrary locations and at arbitrary moments, and thus the need to frequently rewrite the magnetic pattern on the magnetic sheet is kept to a minimum.

2.1 Types of Haptic Feedback

Types of haptic feedback our method presents are bumpy sensations and vibration, with the advantage of seamlessly switching between these senses using a simple method of electric current control. The technique of generating bumps when a specific pattern of horizontal pressure is applied to the user's manipulated object has been studied before [8, 13]; there has also been a study that demonstrates the presentation of bumps using repulsive and attractive magnetic forces [3]. Thus, we can present a bumpy feeling simply by placing a magnetized sheet and moving a tangible with an embedded electromagnet horizontally across the surface of the sheet. Furthermore, the sense of vibration can be presented by switching the direction of the current to the electromagnet periodically at high speed.

Bumpy sensations. The main haptic stimulus presented in this study is a feeling of bumpiness (see Figure 1 (b)). When the magnetic sheet and electromagnet alternately generate attractive and repulsive forces with each other, a feeling of bumpiness is presented. By magnetizing the sheet so that the polarity alternates and then simply energizing the electromagnet, a bumpy texture can be presented on the touch device. Also, the resolution of the bumpiness can be changed dynamically while the tangible is in use by periodically switching the direction of current to the electromagnet.

Vibration. The periodic switching of the current direction to the electromagnet can also be performed at high speed to present a sense of vibration (see Figure 1 (b)). This method of switching the direction of direct current at high speed can be considered equivalent to alternating current, and a similar sensation is produced to that of the vibration of a speaker cone. As an example, in a straightforward calculation, switching the current every 20 ms, a vibration of approximately 23 Hz is presented².

2.2 System Setup

For the magnetic sheet under the device, we used sheets of 3 mm thickness³ to produce sufficient magnetic force, and it was magnetized with a pitch of 10 – 13 mm. We prepared a touch device with a thickness of 4 mm⁴ and an electromagnet with an overall diameter of 19 mm, containing an iron core approximately 6 mm

in diameter, with a coil of 246 turns⁵. We applied approximately 6 V and the current flowing was 1.2 A. The magnetic flux density appearing on the surface of the iron core was about 90 mT.

To dynamically present the multiple haptic feedback during tangible interaction, we implemented a system to control the energization of the electromagnet according to the tangible's position on the touchscreen. The position of the tangible during manipulation can be acquired from the touch position on the screen by placing a conductive material from the bottom of it to the position that the user's fingers contact it. The touch position is obtained as `mouseX` and `mouseY` from the running screen of Processing software displayed on the touchscreen. Then, depending on these positions, Arduino controls the H-bridge circuit⁶ and switches whether and in which direction the electromagnet is energized.

3 APPLICATIONS

Multi-functional Controller. The proposed system can be used with multiple controllers, such as a slider with bumps, a button with a click feeling when pressed, and a thumbstick with a virtual spring sensation without modifying a single module (see Figure 2 (a)). To realize the slider that presents bumps, a stripe pattern is magnetized onto a magnetic sheet, and simply initiating energization is required. Also, by switching the current, it is possible to provide a finer bumpy sensation according to a finer scale slider without changing the magnetization pattern of the sheet. To realize a button with a click feeling, a magnetic sheet should be magnetized only where the button is to be located on the screen. When a touch is detected in the button area, fast energization switching (with 80 ms alternations) is performed only once. As for the thumbstick, the magnetic sheet should be magnetized only at the center of the initial placement location. By energizing the electromagnet so that its polarity is opposite to the magnetized polarity, a thumbstick returns to its initial position after being manipulated. We achieved this by implementing a program that switches the energizing pattern to that of each controller according to the position of the touch input (see Figure 3).

Drawing Tool. We implemented a paint tool with multiple drawing modes such as a brush, a pencil, and a piece of chalk (see Figure 2 (b)). The magnetic sheet was magnetized in a striped or checker pattern. When selecting the brush mode, the user feels no haptic stimuli while drawing. On the other hand, selecting the pencil mode provides a fine bumpy feeling while drawing, and selecting the chalk mode provides a larger bumpy feeling while drawing. These modes can be switched continuously while using the same single module and magnetized sheet through the appropriate switching of the current to the electromagnet.

Interactive Game. We implemented a maze game application that provides interactive haptic feedback (see Figure 2 (c)). The magnetic sheet was magnetized in a striped or checker pattern. The user grasps the tangible and plays a maze game displayed on the touch screen. When detecting that the tangible manipulated by the user has made contact with one of the walls of the maze, the electromagnet is energized to generate vibrations, giving the

²It takes into account the latency of 6.6 ms required to stabilize the electromagnet due to the reactance of the coil.

³NICHILAY MAGNET CO., LTD., MGS004014C Magnetic sheet.

⁴EHOMEWEI E156DSL Mobile Monitor

⁵Esooho, Gikfun Copper Magnetic Levitation Coil with Iron Core

⁶We used an IC module containing two TPC8407 for the H-bridge IC.: <https://akizukidenshi.com/catalog/g/gK-11338/>

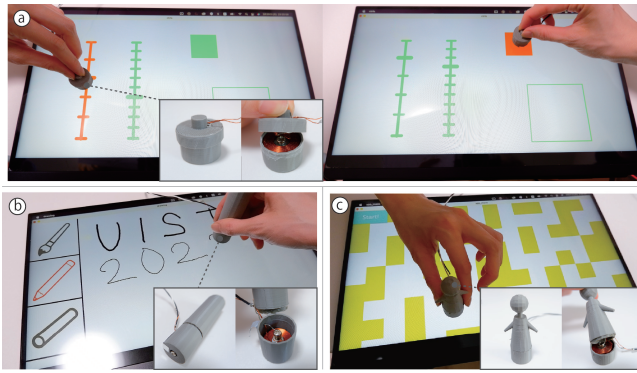


Figure 2: (a) Multi-functional controllers. The user can feel bumps while manipulating a slider (left) and can feel a click feeling when pressing a button (right). **(b) Drawing tool.** The user can feel fine bumps while drawing in pencil mode. **(c) Maze game.** When a tangible touches the walls of the maze, the user can feel a vibration as a sense of damage.

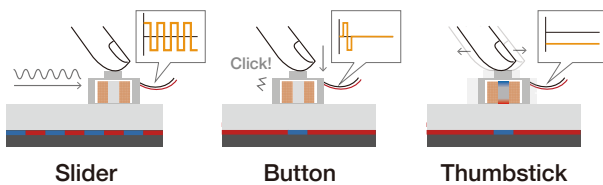


Figure 3: Examples of controllers with haptic feedback.

user a sense of receiving damage from contact with the wall. The game level increases when the user arrives at the goal, and another stage with a higher difficulty level is displayed, in which moving the tangible on the maze floor presents a bumpy feeling. Even in these stages, when the tangible object comes into contact with a wall, the system automatically switches to a vibration sensation to present the sense of receiving damage.

4 CONCLUSION AND FUTURE WORK

We proposed a TUI that, while maintaining a high level of simplicity, dynamically provides bumps and vibrations during tangible interaction on touch devices. We combined a magnetic sheet, which allows easy customization of magnetization patterns, and an electromagnet, which allows control of magnetic forces, and exploited magnetic interference passing through the touch device. Our future work is to investigate the presenting force under different conditions of magnetic materials. We will also develop a software design tool that supports the creation of haptic patterns.

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